

Skills Summary

Expressions: Distribute, Combine, Factor

Use this reference while completing your worksheet or when studying.

1. Distributing over a sum (distribute-over-sum)

Rule: $a(b + c) = ab + ac$ and $a(b - c) = ab - ac$

Why: $a(b + c)$ is the area of a rectangle of height a whose base is cut into b and c . Cutting the base cuts the area into ab and ac . The outside factor multiplies **every** term inside — and each term keeps the sign in front of it.

Example: Expand $6(x + 5)$.

Step 1. Two terms sit inside one set of parentheses, so this is a distribution: multiply the outside factor by each term.

Step 2. $6 \cdot x = 6x$ and $6 \cdot 5 = 30$, so $6(x + 5) = 6x + 30$

Try it: Expand $3(4x + 7)$.

check: $12x + 21$

2. Combining like terms (combine-like-terms)

Like terms have exactly the same variable part: $5x$ and $3x$ are like; $5x$ and 5 are not.

Rule: add or subtract the coefficients and keep the variable part: $5x + 3x = 8x$.

Every term travels with the sign in front of it, so $9 - 4$ is a subtraction of like number terms, not a stray minus.

Example: Simplify $5x + 9 + 3x - 4$.

Step 1. There are no parentheses, only terms of two different kinds, so this is pure sorting: group like with like.

Step 2. $(5x + 3x) + (9 - 4) = 8x + 5$

Try it: Simplify $7x + 12 - 2x - 5$.

check: $5x + 7$

3. Distribute, then combine (distribute-then-combine)

Order matters: you can only combine terms once nothing is trapped inside parentheses.

Step 1 distribute every product. **Step 2** collect like terms.

A term *outside* the parentheses (like the $5x$ in $2(x + 6) + 5x$) is never multiplied by the outside factor — it only joins in at step 2.

Example: Simplify $2(x + 6) + 5x$.

Step 1. Part of this expression is locked inside parentheses, so distribute first and only then look for like terms.

Step 2. $2(x + 6) = 2x + 12$, then $2x + 12 + 5x = (2x + 5x) + 12 = 7x + 12$

Try it: Simplify $4(3x - 1) - 2x$.

check: $10x - 4$

4. Factoring out a common factor

Rule (distribution backwards): $ab + ac = a(b + c)$

Find the **greatest** common factor of the terms, write it outside, and divide each term by it to get what stays inside. Check by distributing: you must land back on the expression you started with.

Example: Factor $8x + 20$.

Step 1. Both terms share a whole-number factor, so run the area model backwards: the shared factor is the height.

Step 2. GCF of 8 and 20 is 4; $8x \div 4 = 2x$ and $20 \div 4 = 5$, so $8x + 20 = 4(2x + 5)$

Try it: Factor $9x - 24$.

check: $(3x - 8)3$

(!) Watch out: $3(2x - 5) \neq 6x - 5$. The outside factor must reach the second term too: $6x - 15$. Test any doubtful step by substituting a number — equivalent expressions agree for every value.